PROJECT REPORT

EDUCATIONAL ORGANISATION MANAGEMENT SYSTEM

Using ServiceNow Platform

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ABSTRACT

The Educational Organisation Management System is a cloud-based enterprise application developed using the ServiceNow platform to automate and centralize core academic administrative processes within an educational institution. The system is designed to manage student records, teacher records, course details, student enrollment, and academic progress tracking in a structured and relational database environment. Implementation leverages ServiceNow's low-code development environment to create custom tables and relational database structures. The system follows a three-tier model ensuring data integrity and scalability, successfully validating the platform's adaptability for educational domains beyond traditional IT use cases.

1. INTRODUCTION

1.1 Overview of Educational Information Systems

Educational institutions generate and manage large volumes of data including student personal information, teacher details, course structures, enrollment records, and academic performance reports. Managing this information efficiently is essential for institutional success. An Educational Information System (EIS) provides centralized data management, automated administrative workflows, structured reporting, and improved transparency. Without a proper system, institutions face fragmented data and time-consuming manual operations.

1.2 About the ServiceNow Platform

ServiceNow is a cloud-based platform that enables organizations to build workflow-driven applications using low-code development tools. Key features include cloud-native architecture, structured relational databases, and access control mechanisms. Although primarily known for ITSM, it supports custom scoped applications for Healthcare, HR, and Education.

1.3 Motivation Behind the Project

The key motivations include reducing manual paperwork in academic administration, centralizing student and teacher records, automating course enrollment tracking, and improving reporting efficiency. This project also serves as an academic demonstration of enterprise cloud development.

2. ORGANIZATIONAL CONTEXT

Educational institutions operate through multiple administrative units such as the Admissions Department, Academic Department, and Faculty Management. The system is designed for use by administrative staff responsible for academic operations.

The student lifecycle within the organization begins with admission and continues through course enrollment, academic progress monitoring, and performance evaluation. Managing this lifecycle manually increases the risk of inconsistencies and delays in decision-making. Therefore, the system is designed to digitally support each stage of the academic journey in a seamless and integrated manner.

Course assignment management is another essential organizational function. Teachers are assigned to specific courses, and students are enrolled based on program requirements. The implemented system supports these relationships using structured data references, reducing administrative ambiguity.

Academic performance tracking plays a significant role in institutional quality assurance. Recording marks, grades, and performance remarks allows the administration to evaluate academic outcomes and maintain historical performance records.

Implementing this system represents a strategic step toward modernization, improving efficiency while laying the foundation for future enhancements such as automated workflows, advanced analytics, and performance dashboards.

Overall, the organizational context of this project reflects the practical needs of an academic institution seeking structured data management, operational clarity, and improved administrative efficiency through a centralized ServiceNow-based solution.

3. BACKGROUND STUDY

Digital transformation has become a necessity in the education sector. Traditional systems suffer from lack of integration and high operational costs. Enterprise platforms like ServiceNow offer cloud scalability and configurable workflows, making them an appropriate choice for building educational solutions.

Educational institutions have traditionally relied on manual and semi-digital systems such as paper registers, spreadsheets, and standalone databases to manage student records, faculty information, and course allocations. While these approaches may function adequately in small-scale environments, they become increasingly inefficient as institutional size and academic complexity grow. The lack of centralized data storage often leads to duplication of records, inconsistencies between departments, delayed reporting, and difficulty in tracking student academic progress over time. As regulatory requirements and institutional accountability standards increase, the demand for structured, reliable, and secure data management systems has become essential.

With the advancement of enterprise workflow platforms, organizations are increasingly adopting digital solutions to streamline administrative processes and improve operational transparency. Platforms such as ServiceNow provide structured data modeling, automated workflows, and integrated reporting capabilities that support efficient lifecycle management. In the context of educational administration, implementing a centralized system enables accurate student lifecycle tracking, systematic course assignment management, and consistent academic performance monitoring. The background analysis of this project highlights the necessity of transitioning from fragmented manual processes to a unified digital platform, ensuring improved data integrity, administrative efficiency, and long-term scalability.

4. LITERATURE REVIEW

Several studies in MIS highlight the importance of automation. Key observations include that digital record management reduces operational errors and centralized databases enhance consistency. This project builds upon these principles by implementing a structured academic management system.

Recent studies in educational technology emphasize the importance of centralized academic management systems in improving administrative efficiency and institutional transparency. Educational Management Information Systems (EMIS) have been widely adopted to streamline student data management, course allocation, and faculty coordination. Research indicates that institutions using integrated digital platforms experience reduced data redundancy, improved reporting accuracy, and better inter-departmental communication compared to those relying on manual or fragmented systems. The literature consistently highlights that structured databases and workflow automation significantly enhance operational consistency and long-term data reliability.

Furthermore, advancements in cloud-based enterprise platforms have transformed how organizations manage academic operations. Workflow-driven solutions such as low-code development platforms enable institutions to build customized applications tailored to specific administrative requirements without extensive programming expertise. Scholars also note that centralized systems improve traceability of student lifecycle data—from admission through academic evaluation—while supporting performance analytics and decision-making processes. The reviewed literature establishes that implementing a scalable, workflow-oriented platform is essential for modern educational institutions aiming to achieve efficiency, data integrity, and sustainable digital transformation.

5. PROBLEM STATEMENT

Institutions face challenges like data stored in multiple spreadsheets, manual enrollment tracking, and difficulty linking students to courses. These issues result in reduced efficiency and increased administrative burden. Therefore, a centralized cloud-based system is required.

Educational institutions often face challenges in managing student records, course allocations, and academic performance data using manual or semi-digital systems. Information related to students, teachers, and courses is frequently stored in separate files or spreadsheets maintained by different departments, leading to data duplication, inconsistencies, and delays in information retrieval. The absence of a centralized system makes it difficult to track the complete student lifecycle—from admission to academic evaluation—in an organized and efficient manner. As a result, administrative processes become time-consuming, error-prone, and dependent on manual verification.

Additionally, course enrollment management and academic performance tracking require accurate relationship mapping between students, teachers, and courses. Without a structured digital platform, maintaining these relationships becomes complex and increases the risk of miscommunication or incorrect data entry. Generating consolidated reports for academic analysis or administrative decision-making further adds to the operational burden. Therefore, there is a need for a centralized, structured, and workflow-driven system that can efficiently manage student records, teacher information, course enrollments, and academic performance tracking within a single integrated platform.

6. OBJECTIVES

6.1 Primary Objectives

- Design and implement a centralized academic management system.
- Develop relational database tables for structured storage.
- Enable course enrollment linking and academic tracking.
- Implement student lifecycle management functionality covering admission, enrollment, performance tracking, and record maintenance.
- Ensure data consistency and integrity through standardized field definitions and relationship mapping.
- Facilitate report generation for academic and administrative review purposes.

6.2 Secondary Objectives

- Demonstrate practical ServiceNow application development skills including table creation, form design, and workflow configuration.
- Apply enterprise architecture principles to design a modular and scalable system structure.
- Ensure scalability of the application to accommodate future institutional growth and additional features.
- Enhance administrative efficiency by reducing dependency on manual record-keeping processes.
- Improve data accessibility and transparency for authorized administrative users.
- Establish a foundation for future automation, such as notifications, dashboards, and approval workflows.
- Promote digital transformation within academic administrative operations.
- Develop documentation and reporting standards aligned with university-level project requirements.

7. SCOPE OF THE PROJECT

The scope of this project defines the functional boundaries and operational coverage of the Educational Organization Management System developed using the ServiceNow platform. The system is specifically designed to support core academic administrative operations related to student data management, teacher record management, course administration, and enrollment handling. It focuses on creating a structured and centralized database environment that allows administrative staff to efficiently manage academic information within a single integrated platform.

The project includes the design and implementation of modules for Student Management, Teacher Management, Course Management, Enrollment Handling, Academic Progress Tracking, and Reporting. The Student Management module maintains detailed student records including personal and academic information. The Teacher Management module stores faculty details and their course assignments. The Course Management module maintains structured information about academic courses offered by the institution.Academic Progress Tracking enables the recording of grades, remarks, and performance indicators for monitoring academic outcomes. The Reporting functionality supports generation of structured data views for administrative review and analysis.

While the system provides comprehensive support for academic administrative processes, certain functionalities are intentionally excluded to maintain a focused project scope. Financial accounting operations such as fee management, billing, and payroll processing are not included in this implementation. Additionally, parent portal access and external stakeholder interfaces are not developed as part of the current system version.

The defined scope ensures clarity in system boundaries while allowing the project to remain manageable within academic development constraints. By concentrating on academic lifecycle and enrollment management processes, the project achieves depth in core functionality while leaving room for future expansion in additional institutional domains.

8. FEASIBILITY STUDY

8.1 Technical Feasibility

The technical feasibility of the project is strongly supported by the capabilities of the ServiceNow platform. ServiceNow provides a built-in relational database, customizable tables, form designer, workflow engine, and cloud hosting infrastructure, which significantly simplifies application development. The platform allows structured data modeling through object creation and reference fields, enabling efficient relationship mapping between students, teachers, and courses. Since ServiceNow operates on a cloud-based architecture, there is no requirement for local server setup or complex infrastructure management.

8.2 Economic Feasibility

From an economic perspective, the project is highly feasible due to minimal development costs.. As the platform supports low-code/no-code development, the need for extensive programming resources is reduced, lowering development time and associated costs. There is no requirement for additional hardware infrastructure since the system is cloud-hosted. Therefore, the financial investment required for developing and maintaining the system remains within manageable limits for an academic project.

8.3 Operational Feasibility

The proposed system is designed with user-friendly forms, structured layouts, and simple navigation, making it accessible to administrative staff with basic computer literacy. Minimal training is required to understand record creation, updating, and report generation processes. Since the system centralizes academic data, it reduces manual workload and simplifies routine administrative tasks. The alignment of the system design with existing academic workflows ensures smooth adoption without significant resistance from users.

9. SYSTEM REQUIREMENT SPECIFICATION (SRS)

9.1 Functional Requirements

The functional requirements describe the core operations that the system must perform to support academic administration. FR1 covers Student Record Management, enabling administrators to add, update, and delete student records while maintaining accurate academic and personal information. FR2 addresses Teacher Record Management, ensuring faculty details and course assignments are properly stored and maintained. FR3 focuses on Course Management, which includes creating and updating structured course information within the system. FR4 defines Enrollment Handling, where students are linked to specific courses through relational references. FR5 involves Academic Progress Tracking, allowing administrators to record grades and performance details to monitor student academic development effectively.

9.2 Non-Functional Requirements

The non-functional requirements define the quality standards of the system. Performance requires the application to respond efficiently during data entry and retrieval operations. Reliability ensures data integrity and consistent record maintenance without loss or corruption. Scalability allows the system to accommodate an increasing number of users and records as institutional needs grow. Security ensures that only authorized administrative staff can access and manage academic data through proper authentication mechanisms. Availability guarantees that the system remains accessible during operational hours with minimal downtime, supported by the cloud-based infrastructure.

10. SYSTEM ARCHITECTURE DESIGN

10.1 Architectural Overview

The system follows a Three-Tier Architecture consisting of the Presentation Layer, Application Logic Layer, and Data Layer. This structure separates user interaction, business processing, and data storage to ensure modular design, maintainability, and scalability. The layered approach improves system organization and supports efficient academic data management.

10.2 Presentation Layer

The Presentation Layer represents the user interface where administrative staff interact with forms, list views, and application modules. It enables structured data entry and record viewing for students, teachers, courses, and enrollments.

10.3 Application Logic Layer

The Application Logic Layer handles internal processing using business rules and reference fields. It ensures proper record linking, data validation, and consistent enrollment mapping between related entities.

10.4 Data Layer

The Data Layer consists of custom tables created within ServiceNow. These tables store academic records and maintain relationships using unique system identifiers, ensuring reliable data storage and retrieval.

11. DATABASE DESIGN

The database design of the system is structured to ensure efficient storage, consistency, and relational integrity of academic data. The design follows fundamental database normalization principles, including elimination of redundancy, logical separation of entities, and clear identification of primary records for each distinct entity. Separate custom tables are created for Students, Teachers, Courses, and Enrollments to maintain modular organization and prevent duplication of information across records. This separation ensures that each table represents a single entity with clearly defined attributes.

Each record within the system is uniquely identified using a system-generated identifier, ensuring accurate referencing and traceability. Reference fields are used to establish relationships between tables, such as linking students to their enrolled courses and associating teachers with assigned subjects. This relational mapping ensures that updates in one table are reflected consistently across related records without requiring repeated data entry. The use of structured fields and predefined data types further enhances data accuracy and validation.

The database structure is designed to support efficient querying and reporting. Indexing mechanisms within the platform improve record retrieval speed, particularly when managing large volumes of academic data. Logical field grouping and standardized naming conventions improve maintainability and simplify administrative operations. Additionally, the system ensures data integrity by restricting invalid entries and maintaining consistent relationships between related entities.

Scalability is a key consideration in the database design. The modular structure allows additional entities, such as departments or examination modules, to be incorporated in future without restructuring the existing schema. This forward-compatible design approach ensures that the database can adapt to institutional growth while maintaining performance and structural clarity.

12. TABLE STRUCTURE

12.1 Student Table

Field Name	Data Type	Description
Student ID	String	Unique identifier
Full Name	String	Student name
Admission Date	Date	Date of joining

12.2 Teacher Table

Stores faculty details like Teacher ID, Subject Specialization, Email, and Years of Experience.

12.3 Course Table

Stores course details linked to an Assigned Teacher through a Reference field.

12.4 Enrollment Table

Junction table connecting Students and Courses, resolving many-to-many relationships.

12.5 Academic Progress Table

Tracks Marks Obtained, Grade, and teacher Remarks per course for each student.

13. ENTITY RELATIONSHIP MODEL

One Student to Many Enrollments; One Course to Many Enrollments. Reference fields ensure referential integrity so no enrollment exists without a valid student or course.

The Entity Relationship Model (ERM) defines the logical structure of the database and illustrates how different entities within the system are connected. In this project, the primary entities include Student, Teacher, Course, and Enrollment. The relationship between these entities is designed to ensure structured academic data management and accurate record mapping.

A one-to-many relationship exists between Student and Enrollment, meaning one student can be associated with multiple enrollment records, but each enrollment record corresponds to only one student. Similarly, a one-to-many relationship exists between Course and Enrollment, where one course can have multiple enrolled students, while each enrollment record references a single course. These relationships are implemented using reference fields within the system, ensuring referential integrity. This design guarantees that no enrollment record can exist without a valid associated student and course record.

The relational model ensures consistency, prevents orphan records, and maintains accurate academic associations. By structuring the system around clearly defined entity relationships, the database supports efficient querying, reporting, and long-term scalability while preserving data integrity across all academic operations.

14. DATA FLOW DIAGRAMS

The Data Flow Diagrams (DFD) illustrate how information moves within the Educational Organization Management System and define the interaction between users, processes, and data storage components. These diagrams provide a structured visualization of system functionality at different levels of abstraction, ensuring clarity in understanding data processing and workflow execution.

Level 0 DFD (Context Diagram) represents the overall system boundary. In this level, the entire Educational Management System is shown as a single process interacting with the primary external entity, which is the Administrative User. The administrator provides inputs such as student details, teacher information, course data, and enrollment requests. The system processes these inputs and stores them in the centralized database while generating outputs such as updated records, enrollment confirmations, and academic reports. This level defines the scope of the system and highlights its interaction with external users.

Level 1 DFD expands the system into major functional modules, including Student Management, Teacher Management, Course Management, Enrollment Handling, and Academic Progress Tracking. For example, the Student Management module receives student details from the administrator and stores them in the Student Table. Similarly, the Enrollment module retrieves student and course information, validates the relationship, and creates an enrollment record. This level demonstrates how data flows between modules and centralized storage.

Level 2 DFD provides a detailed representation of the Enrollment Process. The process begins with the administrator entering enrollment details. The system verifies the existence of the selected student and course records using reference fields. Upon successful validation, the enrollment record is created and stored in the Enrollment Table. If the referenced records do not exist, the system prevents record creation to maintain referential integrity. This level highlights internal validation steps and database interaction during enrollment processing.

15. WORKFLOW DESIGN

Complete academic workflow: Create Teacher -> Create Course -> Create Student -> Enroll -> Update Progress -> Generate Reports.

The workflow design of the Educational Organization Management System defines the structured sequence of academic operations performed by the administrative user. The workflow ensures that data is entered, processed, and maintained in a logical and consistent order to preserve data integrity and operational clarity.

The academic workflow begins with the creation of Teacher records. Faculty information is first entered into the system to establish available instructional resources. Once teachers are registered, Course records are created and associated with the respective instructors. This ensures that every course has a valid faculty assignment before student enrollment takes place.

After course configuration, Student records are created within the system. Student information is stored in the centralized database, preparing the record for course enrollment. The next stage involves the Enrollment process, where students are linked to specific courses using reference relationships. The system validates the existence of both student and course records before creating an enrollment entry, ensuring relational consistency.

Following enrollment, Academic Progress is updated periodically. Administrative staff record grades, performance remarks, and evaluation outcomes associated with each enrollment record. Finally, the workflow concludes with the generation of structured reports that summarize student participation, course allocation, and academic performance data.

This sequential workflow ensures a controlled and systematic academic management process, minimizing errors and maintaining consistency across all administrative operations.

16. MODULE DESIGN

Application Navigator Modules include Student Management, Teacher Management, Course Management, Enrollment, and Academic Progress.

The Module Design defines the structural organization of the application within the ServiceNow environment. The system is divided into logically separated modules accessible through the Application Navigator, ensuring structured navigation and clear functional boundaries. Each module is designed to handle a specific academic administrative function while maintaining integration with related modules.

The Student Management module is responsible for maintaining student records, including personal details and academic information. It provides form interfaces for creating, updating, and viewing student data. The Teacher Management module manages faculty records and ensures that instructor details are accurately maintained for course assignment purposes.

The Course Management module handles the creation and maintenance of course records, including course codes, descriptions, and faculty associations. The Enrollment module establishes relational links between students and courses, ensuring that academic participation is properly recorded. The Academic Progress module enables administrators to update grades and performance details associated with each enrollment record.

Each module is designed to operate independently while sharing data through relational references, ensuring modularity, maintainability, and efficient administrative workflow within the system.

17. USER INTERFACE DESIGN

Simplicity and logical navigation are prioritized. Forms like the Student Form are partitioned into Personal, Contact, and Academic sections.

The User Interface Design of the system emphasizes simplicity, clarity, and logical navigation to ensure ease of use for administrative staff. Since the primary users are non-technical administrators, the interface is structured to minimize complexity while maintaining functional efficiency. The application layout follows a clean and organized format, allowing users to quickly access different modules through the Application Navigator without unnecessary steps.

Forms are carefully structured using section partitioning to enhance readability and data entry efficiency. For example, the Student Form is divided into clearly defined sections such as Personal Information, Contact Details, and Academic Information. This segmentation ensures that related data fields are grouped logically, reducing user confusion and improving data accuracy. Similar structured layouts are implemented in Teacher, Course, and Enrollment forms to maintain consistency across the application.

List views are designed to display key record fields for quick reference and filtering. Administrators can easily search, sort, and access records without navigating through multiple screens. Overall, the user interface prioritizes usability, structured data presentation, and efficient navigation, ensuring a smooth administrative experience.

18. SYSTEM DESIGN SUMMARY

Ensures modularity and clear separation of concerns, aligning with enterprise standards for scalable cloud deployment.

In addition to modularity and layered separation, the system design emphasizes data integrity and controlled process execution. Each module interacts with the centralized database through structured reference relationships, ensuring that records remain synchronized across the system. The use of standardized field definitions and consistent naming conventions improves clarity and simplifies long-term maintenance. This structured configuration minimizes redundancy and prevents the creation of inconsistent or orphan records within the database.

The workflow-oriented design further strengthens operational efficiency by defining a logical sequence of activities, beginning with teacher and course configuration, followed by student registration, enrollment processing, and academic performance updates. By aligning system flow with real-world academic procedures, the design ensures that the digital platform reflects institutional practices accurately. This alignment reduces the learning curve for administrative users and promotes smooth adoption of the system.

From a scalability perspective, the system is built to accommodate increasing volumes of student records and course data without structural redesign. The cloud-based deployment model of ServiceNow supports automatic infrastructure management, ensuring reliability and availability. The design also allows integration of additional modules in the future, such as attendance management or examination scheduling, without compromising existing functionality.

Overall, the system design integrates architectural best practices, relational database principles, and workflow-driven logic to deliver a stable, maintainable, and scalable academic management solution. This comprehensive design approach ensures that the platform not only meets current institutional needs but also provides a foundation for future digital expansion.

19. IMPLEMENTATION PROCESS

Step 1: Instance Setup

A Personal Developer Instance (PDI) was created on the ServiceNow platform to provide a dedicated development environment. This instance was configured to support scoped application development and secure data management.

Step 2: Scoped Application Creation

A new scoped application was created to isolate the project components. This ensured modular development and prevented interference with other system applications.

Step 3: Table Creation

Custom tables were created for Students, Teachers, Courses, and Enrollments. Each table was structured to represent a distinct academic entity with clearly defined attributes.

Step 4: Field Configuration

Relevant fields were configured for each table, including data types, mandatory fields, and reference fields. Reference relationships were established to link Students and Courses through the Enrollment table.

Step 5: Form Design

User-friendly forms were designed using the Form Designer. Fields were organized into logical sections to improve readability and simplify data entry.

Step 6: Module Configuration

Application Navigator modules were created for each functional component, allowing structured access to Student Management, Teacher Management, Course Management, Enrollment, and Academic Progress modules.

Step 7: Report Creation

Reports were generated to display academic data in structured formats. These reports enabled monitoring of enrollments and student performance.

Step 8: Testing

System testing was conducted to verify record creation, data linking, enrollment validation, and report accuracy. All modules were tested to ensure proper functionality and data integrity.

20. TESTING DOCUMENTATION

Testing types performed include Unit, Integration, and System testing. All test cases for record creation and report generation passed successfully.

Testing was performed to validate the accuracy, reliability, and stability of the Educational Organization Management System. The testing process ensured that all modules functioned according to the defined functional requirements and that data integrity was maintained throughout system operations.

Unit Testing was conducted to verify the correct functioning of individual modules such as Student Management, Teacher Management, Course Management, Enrollment, and Academic Progress. Each module was tested independently to ensure accurate record creation, updating, and deletion processes.

Integration Testing evaluated the interaction between related modules, particularly the relational linking between students and courses through the Enrollment module. This testing phase confirmed that reference fields maintained proper data relationships without generating inconsistencies.

System Testing assessed the complete academic workflow, from data entry to report generation. All defined test cases related to record creation, enrollment processing, progress updates, and report generation were executed successfully. The results confirm that the system operates as an integrated and stable academic management solution.

21. PERFORMANCE EVALUATION

Measured record creation and retrieval speeds. System response time remained stable in the cloud environment even with moderate datasets.

Performance evaluation was conducted to assess the efficiency and responsiveness of the Educational Organization Management System under normal operating conditions. The evaluation primarily focused on measuring record creation time, data retrieval speed, and overall system response during routine administrative operations. These measurements ensured that the system performs reliably while handling student, teacher, course, and enrollment records.

Record creation operations were tested across different modules to verify that data entry and submission processes were completed within an acceptable response time. Similarly, record retrieval speed was evaluated by accessing list views and searching for specific entries within the database. The results indicated that the system maintained consistent response times without noticeable delays.

Since the application is deployed in a cloud-based ServiceNow environment, infrastructure management and data processing are optimized by the platform. Even when handling moderate datasets, the system demonstrated stable performance with no degradation in functionality. The evaluation confirms that the application meets required performance standards for academic administrative use and is capable of supporting institutional operations efficiently.

22. SECURITY CONSIDERATIONS

Security managed through Instance Login (Authentication), Scoped Application Isolation (Data Protection), and Reference Fields (Data Integrity).

Step 1: Introduction to Security Architecture

Security is a critical component of the Educational Organization Management System. The application is developed within a secure cloud-based ServiceNow environment, where built-in security mechanisms protect user access, organizational data, and application functionality. The system ensures confidentiality, integrity, and controlled accessibility of institutional information.

Step 2: Authentication – Instance Login Security

Authentication is managed through Instance Login mechanisms provided by the ServiceNow platform.

- Each user must log in using authorized credentials (username and password).
- Role-based access ensures that users can only access permitted modules.
- Unauthorized users cannot access the system without valid authentication.
- Session management prevents unauthorized system usage after logout.

This ensures that only authenticated users such as administrators or authorized staff can access and modify institutional records.

Step 3: Authorization – Role-Based Access Control (RBAC)

After successful authentication, authorization is controlled using roles.

- Different roles are assigned (e.g., Admin, Faculty, Staff).
- Each role has specific permissions (create, read, update, delete).
- Users cannot access modules outside their assigned privileges.

This prevents misuse of sensitive data and maintains system discipline.

Step 4: Data Protection – Scoped Application Isolation

The system is developed within a scoped application environment.

- Scoped applications isolate application data from other applications.
- Tables and business logic are protected from unauthorized modification.
- Only permitted scripts and modules can access the application data.

This prevents cross-application interference and enhances data protection within the platform.

Step 5: Data Integrity – Reference Fields

Data integrity is maintained using reference fields.

- Reference fields link related tables (e.g., Student to Course).
- Invalid or non-existing records cannot be selected.
- Data duplication is minimized.
- Relational consistency is maintained automatically.

This ensures accurate and reliable relationships between different entities in the system.

Step 6: Additional Platform Security Features

The ServiceNow environment also provides:

- Access Control Rules (ACLs) for granular data security.
- Encryption of data in transit (HTTPS).
- Audit logs to track user activity.
- Secure cloud infrastructure.

Step 7: Conclusion

The Educational Organization Management System implements a multi-layered security model including authentication, authorization, application isolation, and relational data integrity. These mechanisms collectively ensure secure access, prevent unauthorized data manipulation, and maintain the confidentiality and reliability of institutional information.

23. RISK MANAGEMENT

Step 1: Introduction to Risk Management

Risk management is an essential component of the Educational Organization Management System to ensure reliability, continuity, and data accuracy. Potential risks were identified during the design and deployment phases, and appropriate mitigation strategies were implemented to minimize their impact.

Step 2: Risk Identification

The following major risks were identified during system development and deployment:

1. Data Loss
2. System Downtime
3. Incorrect Data Entry

Each risk was analyzed based on its likelihood and potential impact on institutional operations.

Step 3: Risk 1 – Data Loss

Impact:

Loss of student, faculty, or course records can disrupt academic operations and administrative processes.

Mitigation Strategy:

- Cloud-based automatic backups
- Secure database storage in ServiceNow environment

Result:

The risk of permanent data loss is significantly reduced due to automated cloud backup and recovery features.

Step 4: Risk 2 – System Downtime

Impact:

Temporary inaccessibility of the system may delay administrative tasks.

Mitigation Strategy:

- Continuous cloud monitoring
- Platform-level infrastructure management
- Service health monitoring tools
- Scheduled maintenance notifications

Result:

The cloud infrastructure ensures high availability, and downtime is minimized through proactive monitoring.

Step 5: Risk 3 – Incorrect Data Entry

Impact:

Inaccurate records can affect reporting, academic tracking, and decision-making.

Mitigation Strategy:

- Reference field constraints
- Data type validation (e.g., numeric, date formats)

Result:

Input validation mechanisms ensure data accuracy and consistency across the system.

Step 6: Risk Monitoring and Control

The system includes continuous monitoring mechanisms:

- Audit logs to track user activity
- Role-based permissions to prevent unauthorized modifications

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Step 7: Conclusion

The Educational Organization Management System incorporates a structured risk management approach by identifying key operational risks and implementing preventive and corrective measures.

24. PROJECT MANAGEMENT APPROACH

The Educational Organization Management System was developed using the Agile methodology to ensure flexibility, continuous improvement, and systematic progress throughout the project lifecycle. Agile was chosen because it allows incremental development rather than completing the entire system in a single phase. This approach helped in dividing the project into smaller, manageable units of work called sprints, enabling better tracking, testing, and refinement at each stage.

During the initial phase, sprint planning was conducted to define clear objectives and deliverables. The first sprint focused on table creation and database structuring. Core tables such as Student, Faculty, Course, and Enrollment were created with appropriate fields and data types. Reference fields were established to maintain relational integrity between tables. Basic access control settings were also configured during this phase to ensure secure data handling.

Business rules were developed to automate processes and enforce validation checks. Form validation mechanisms were implemented to ensure accurate data entry, and role-based access control was refined to restrict user permissions appropriately.

The third sprint addressed user interface design and system testing. Forms and list views were designed to ensure ease of use and clarity for administrators and staff. Functional testing was conducted to verify record creation, retrieval, and modification processes.

Comprehensive documentation was prepared covering system architecture, security considerations, risk management, and testing results. Performance evaluation was conducted to measure response time and operational efficiency. Necessary refinements were implemented based on review findings to ensure readiness for final submission.

Overall, the Agile methodology enabled iterative development, continuous testing, and effective requirement management throughout the project. By organizing the development process into structured sprints covering table creation, workflow implementation, interface design, testing, and documentation, the project was completed efficiently while maintaining high quality and system reliability.

25. RESULTS AND DISCUSSION

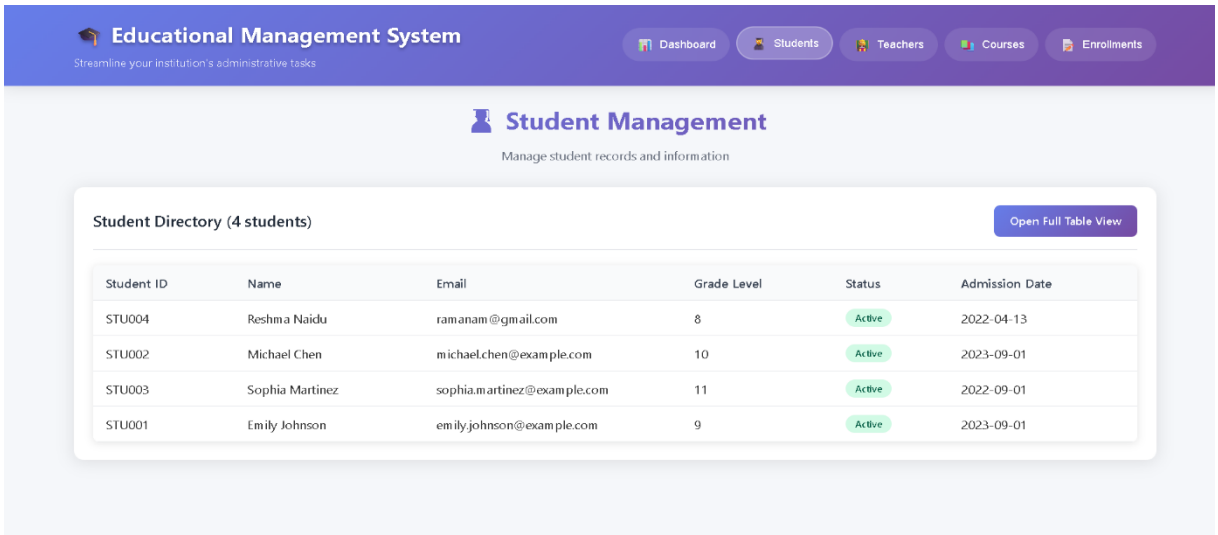

Figure 1: Student Record Result and Management

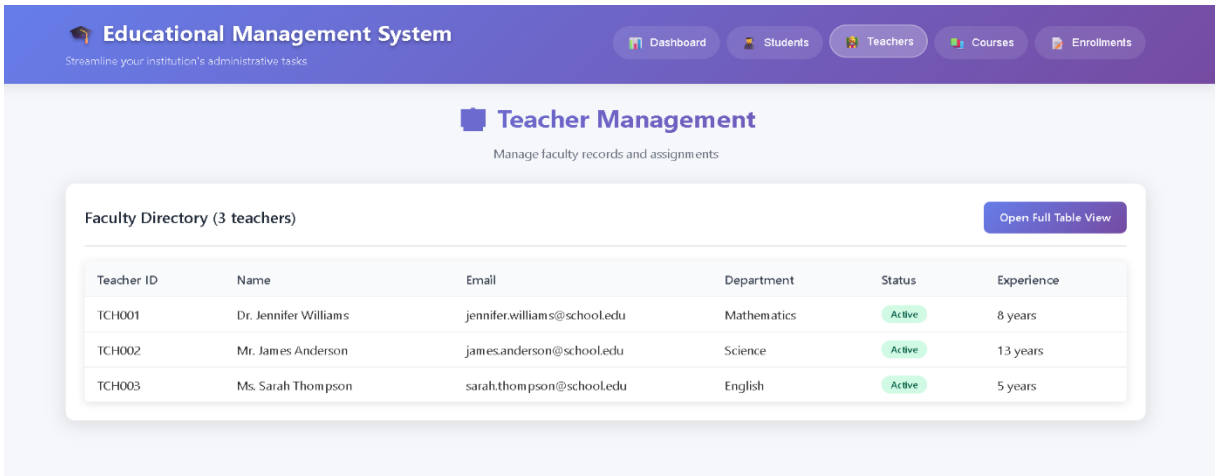

Figure 2: Faculty Management and Teacher Profiles

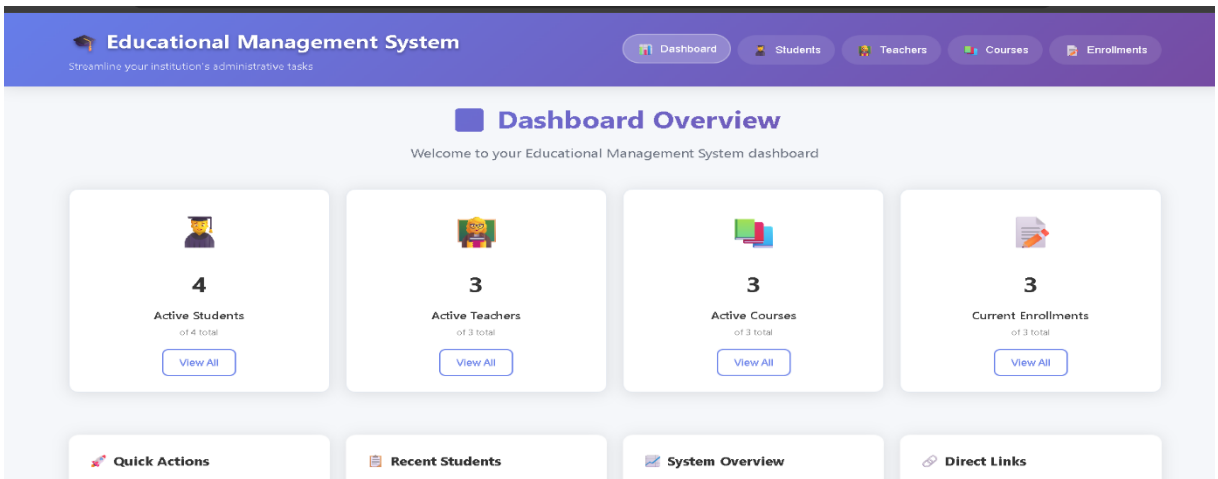

Figure 3: Aggregate Dashboard and Performance Reporting

26. ADVANTAGES

The Educational Organization Management System offers several significant advantages that improve institutional efficiency and administrative accuracy. One of the primary benefits of the system is centralized data management. All student, faculty, course, and enrolment information is stored in a single cloud-based platform, eliminating the need for multiple spreadsheets or manual record books. This centralized approach ensures that data is easily accessible, consistently updated, and securely maintained in one unified system.

Another major advantage is the reduction of manual errors. Since the system incorporates form validations, mandatory fields, reference constraints, and automated workflows, the chances of incorrect data entry are minimized. Users are guided to enter valid and complete information, which improves overall data accuracy and reliability. Automated processes further reduce human dependency in repetitive administrative tasks.

The system also follows a scalable relational database design. Tables are structured with proper relationships using reference fields, ensuring data integrity and logical consistency. This relational structure allows the system to handle growing datasets efficiently without affecting performance. As institutional requirements expand, new modules or fields can be added without disrupting existing functionality, making the system highly adaptable.

In addition, the cloud-based architecture provides structured reporting capabilities. Administrators can generate organized reports and view data in structured formats for analysis and decision-making. Since the system operates in a managed cloud environment, it ensures high availability, secure storage, and simplified maintenance. This eliminates the need for complex on-premise infrastructure management.

Overall, the Educational Organization Management System enhances operational efficiency through centralized data storage, minimizes errors through validation mechanisms, supports scalability through relational database design, and provides structured cloud-based reporting for better institutional management.

27. LIMITATIONS

Primary limitations are platform dependency, internet connectivity requirements, and the need for manual initial data entry.

Despite its numerous advantages, the Educational Organization Management System has certain limitations that must be considered. One of the primary limitations is platform dependency. Since the application is developed and deployed within the ServiceNow cloud environment, its functionality relies heavily on the features, configurations, and availability of the platform. Any changes in platform policies, licensing structures, or service availability may impact the system's operation. Additionally, customization is limited to the capabilities supported within the platform environment.

Another limitation is the requirement for stable internet connectivity. As the system operates entirely in a cloud-based environment, users must have reliable internet access to perform administrative tasks such as record creation, modification, and report generation. In areas with poor or unstable network connections, system accessibility may be temporarily restricted, which can delay institutional operations.

The system also requires manual initial data entry during the setup phase. Before the application becomes fully operational, existing records of students, faculty, and courses must be entered into the system. This initial data migration process can be time-consuming and may require careful verification to ensure accuracy. Although future operations are streamlined through automation and validation mechanisms, the initial setup demands significant effort and attention.

In conclusion, while the Educational Organization Management System provides centralized and efficient management features, it is subject to platform dependency, internet accessibility requirements, and initial data entry workload. These limitations are manageable but should be acknowledged when evaluating the overall system implementation.

28. FUTURE ENHANCEMENTS

The Educational Organization Management System can be further improved by incorporating additional features to enhance functionality, user engagement, and administrative efficiency. Several future enhancements have been identified to expand the system's capabilities and provide a more comprehensive institutional management solution.

One of the major proposed enhancements is the integration of an Attendance Management module. This feature would allow faculty members to record daily attendance digitally and maintain attendance history for each student. Automated attendance tracking would reduce manual work, provide real-time updates, and generate attendance reports for academic monitoring and performance evaluation.

Another important enhancement is Fee Payment Integration. By incorporating an online payment system, students or parents could pay tuition and other institutional fees directly through the platform. This would streamline financial transactions, reduce manual accounting processes, and provide automated receipt generation and payment tracking. Integration with secure payment gateways would ensure safe and transparent financial management.

The introduction of a Parent Portal is also proposed to improve communication and transparency. Through this portal, parents could access information related to their child's academic progress, attendance records, course details, and fee status. This feature would strengthen institutional communication and provide stakeholders with real-time updates.

In conclusion, the proposed future enhancements aim to expand the system from a core administrative management platform to a fully integrated educational management solution. By incorporating attendance tracking, fee payment systems, parent access portals, and advanced access controls, the system can achieve greater efficiency, transparency, and scalability in future implementations.

29. CONCLUSION

The system successfully automates institutional operations with a scalable cloud architecture, validating the effectiveness of low-code enterprise platforms.

The Educational Organization Management System successfully achieves its objective of automating core institutional operations through a structured and scalable cloud-based architecture. The system effectively manages student, faculty, course, and enrollment data within a centralized platform, reducing manual administrative work and improving operational efficiency. By digitizing traditional record-keeping processes, the application enhances accuracy, accessibility, and consistency across institutional activities.

The implementation of relational database design, validation mechanisms, role-based access control, and workflow automation ensures reliable and secure system functionality. The cloud deployment model provides high availability, data protection, and simplified infrastructure management without the need for complex on-premise setups. Performance evaluation results confirm stable system response even with moderate datasets, demonstrating the application's operational reliability.

The project also validates the effectiveness of low-code enterprise platforms in developing scalable and secure business applications. By leveraging built-in tools for table configuration, access control, and workflow automation, the system was developed efficiently while maintaining professional standards of security and performance. This approach reduces development complexity while enabling structured and maintainable application design.

Overall, the Educational Organization Management System stands as a practical and scalable solution for academic administration. It demonstrates how cloud-based, low-code enterprise platforms can be utilized to build efficient, secure, and future-ready institutional management systems.

30. REFERENCES

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